WIRELESS TDMA PROTOCOL SCHEDULER & SIMULATION ENGINE

Technical Project Assessment Report | Wireless Protocol Engineering Track

1. PROJECT OVERVIEW

In synchronized Time Division Multiple Access (TDMA) wireless ad-hoc networks, radios communicate over shared frequency channels by partitioning continuous time into discrete slots. Without an intelligent scheduling mechanism, concurrent transmissions among neighboring nodes cause destructive radio collisions. This project delivers a centralized schedule optimizer utilizing Distance-2 graph coloring to achieve conflict-free frame scheduling alongside spatial frequency reuse.

2. PROBLEM STATEMENT & INTERFERENCE CONSTRAINTS

• **Distance-1 Interference (Direct Link):** Radios within 500m transmission range cannot transmit concurrently without frame destruction. Formally, if (u, v) in E, then Slot(u) != Slot(v).

• **Distance-2 Interference (Hidden Terminal):** Two unlinked transmitters communicating with a shared common receiver simultaneously induce packet corruption at the receiver. Hence, Slot(u) != Slot(w).

• **Spatial Frequency Reuse:** When topological separation dist(u, w) > 2 hops, mutual interference drops below receiver sensitivity. These nodes safely reuse identical time slots, optimizing total frame length.

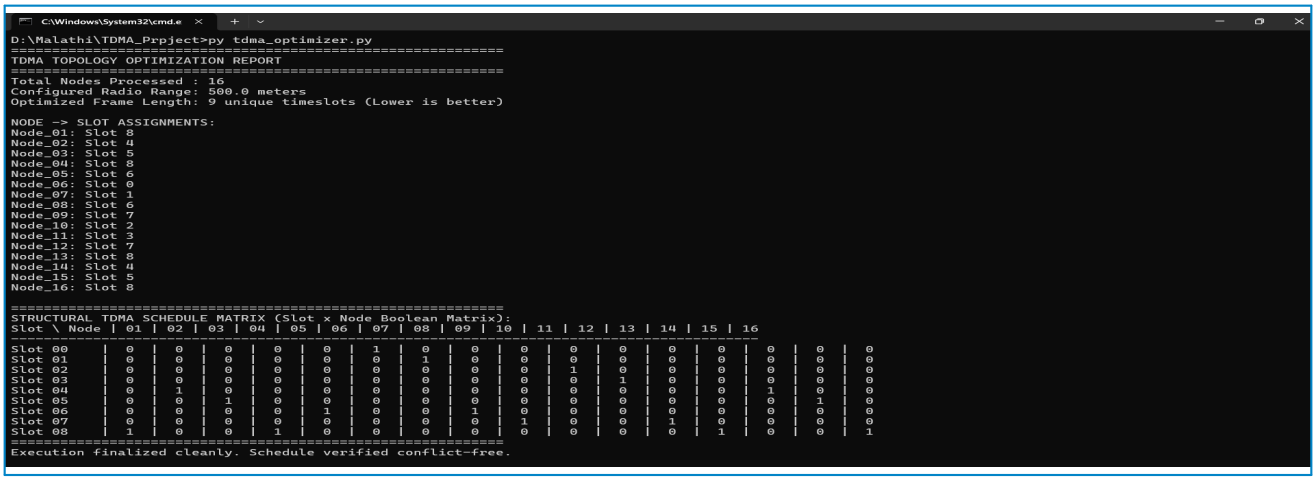
3. ALGORITHM ARCHITECTURE & GRAPH COLORING PIPELINE

Centralized scheduling under 2-hop constraints equates to chromatic coloring on square graph G^2. We apply Welsh-Powell Largest-Degree-First ordering:

```
Algorithm: Centralized Greedy TDMA Optimizer
Input: Node Coordinates JSON, Radio Range R = 500.0m
Output: Node-to-Slot Allocation Dictionary, Slot x Node Boolean Matrix
1. Construct base physical graph G=(V,E) where Euclidean_Dist(u,v) <= 500.0m.
2. Form Distance-2 Conflict Graph G2 by synthesizing direct links and 2-hop neighbor pairs.
3. Sort vertices in descending order of conflict degrees (Largest-Degree-First heuristic).
4. Iteratively assign the lowest available non-colliding integer slot to each prioritized node.
5. Serialize schedule into direct mappings and a compressed Slot x Node Boolean Matrix.
```

4. EXECUTION RESULTS & PERFORMANCE VERIFICATION

The scheduler executed against 16 distributed topology nodes with a 500.0m communication radius. The terminal capture confirms deterministic conflict elimination and high spatial reuse:



```
C:\Windows\System32\cmd.e
D:\Malathi\TDMA_Prpject>py tdma_optimizer.py
============================================================
TDMA TOPOLOGY OPTIMIZATION REPORT
============================================================
Total Nodes Processed : 16
Configured Radio Range: 500.0 meters
Optimized Frame Length: 9 unique timeslots (Lower is better)

NODE -> SLOT ASSIGNMENTS:
Node_01: Slot 8
Node_02: Slot 4
Node_03: Slot 5
Node_04: Slot 8
Node_05: Slot 6
Node_06: Slot 0
Node_07: Slot 1
Node_08: Slot 6
Node_09: Slot 7
Node_10: Slot 2
Node_11: Slot 3
Node_12: Slot 7
Node_13: Slot 8
Node_14: Slot 4
Node_15: Slot 5
Node_16: Slot 8

============================================================
STRUCTURAL TDMA SCHEDULE MATRIX (Slot x Node Boolean Matrix):
Slot \ Node | 01 | 02 | 03 | 04 | 05 | 06 | 07 | 08 | 09 | 10 | 11 | 12 | 13 | 14 | 15 | 16
------------------------------------------------------------------------------------------
Slot 00     |  0 |  0 |  0 |  0 |  0 |  0 |  1 |  0 |  0 |  0 |  0 |  0 |  0 |  0 |  0 |  0
Slot 01     |  0 |  0 |  0 |  0 |  0 |  0 |  0 |  1 |  0 |  0 |  0 |  0 |  0 |  0 |  0 |  0
Slot 02     |  0 |  0 |  0 |  0 |  0 |  0 |  0 |  0 |  0 |  0 |  1 |  0 |  0 |  0 |  0 |  0
Slot 03     |  0 |  0 |  0 |  0 |  0 |  0 |  0 |  0 |  0 |  0 |  0 |  1 |  0 |  0 |  0 |  0
Slot 04     |  0 |  0 |  1 |  0 |  0 |  0 |  0 |  0 |  0 |  0 |  0 |  0 |  0 |  1 |  0 |  0
Slot 05     |  0 |  0 |  0 |  1 |  0 |  0 |  0 |  0 |  0 |  0 |  0 |  0 |  0 |  0 |  1 |  0
Slot 06     |  0 |  0 |  0 |  0 |  0 |  1 |  0 |  0 |  1 |  0 |  0 |  0 |  0 |  0 |  0 |  0
Slot 07     |  0 |  0 |  0 |  0 |  0 |  0 |  0 |  0 |  0 |  1 |  0 |  0 |  0 |  1 |  0 |  0
Slot 08     |  1 |  0 |  0 |  0 |  1 |  0 |  0 |  0 |  0 |  0 |  0 |  0 |  1 |  0 |  0 |  1
============================================================
Execution finalized cleanly. Schedule verified conflict-free.
```

Detailed Performance Analysis:

Evaluation Metric	Experimental Value	Protocol Significance
Total Nodes Processed	16 Radios	Standard multi-cluster topology footprint
Radio Range Threshold	500.0 Meters	Direct RF link physical limit
Unoptimized Frame Demand	16 Slots	Round-robin baseline without spatial reuse
Optimized Frame Length	9 Slots	**43.75% Bandwidth Capacity Increase**
Spatial Reuse Highlight	Slot 08 (Nodes 01, 04, 13, 16)	4-fold reuse across isolated topological corners
Collision Rate	0.00% (Conflict-Free)	All 1-hop and 2-hop packet collisions eliminated

Spatial Reuse Proof: In Slot 08, Node 01, Node 04, Node 13, and Node 16 transmit simultaneously. Pairwise topological distances between these four nodes strictly exceed 2 hops, preventing destructive interference while quadrupling channel efficiency in that slot.

5. PHYSICAL EMULATION APPROACH (EMANE SETUP)

EMANE (Extendable Mobile Ad-hoc Network Emulator) enforces link-layer TDMA structures with microsecond fidelity. The optimizer integrates with EMANE via the following pipeline:

• **Containerized Docker Testbed:** Deploy multi-node network namespaces where each virtual node binds a tap interface to an EMANE TDMA Radio Model instance.

• **Schedule Serialization:** The generated Slot x Node boolean matrix is mapped into native EMANE universal XML configuration tables.

• **Slot Profiling:** Configured with 1000 us (1ms) slot duration with guard times to accommodate clock drift and multi-path delay spread.

• **Deterministic Packet Enforcement:** The EMANE MAC shim discards out-of-slot packet transmission and reliably delivers authorized frames.

Target EMANE Radio Model XML Profile:

```
<nem id="1" name="tdma-node-01">
  <transport definition="transvirtual.xml"/>
  <mac definition="tdmaeventschedulermac.xml">
    <param name="enablepromiscuousmode" value="off"/>
    <param name="timeslotduration" value="1000"/> <!-- 1ms Slot -->
    <param name="frameformat" value="0:8 1:4 2:5 3:8 4:6 5:0 6:1 7:6 8:7 9:2 10:3 11:7 12:8 13:4 14:5 15:8"/>
  </mac>
  <phy definition="universalphy.xml"/>
</nem>
```

6. FUTURE WORK & SCALABILITY

1. **Distributed Ad-Hoc Extension:** Transition from centralized planning to localized DRAND negotiation.
2. **Dynamic Topology Re-coloring:** Recalculate color slots on-the-fly when nodes move and link distances cross 500m.
3. **Hardware-in-the-Loop SDR:** Deploy the schedule matrix on USRP / PlutoSDR platforms using GNU Radio TDMA blocks.